

*Proof.* For the sake of convenience in calculations, we first use Lemma A.5 to replace the Shannon entropy of response distribution.

$$\begin{aligned} \frac{d\gamma_{p\theta}^{\mathcal{D}_j/\mathcal{D}}}{dl} &= \frac{d\left(\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_j} [-p^{\mathcal{D}_j} \log p^{\mathcal{D}}] - \log M\right)}{dl} \\ &= \frac{d\left(\mathbb{E}_{X_{\mathcal{D}_j} \sim \mathcal{P}_{mass}^j, X_{\mathcal{D}} \sim \mathcal{P}_{mass}} [-X_{\mathcal{D}_j} \log X_{\mathcal{D}}]\right)}{dl}. \end{aligned}$$

According to Assumption A.7,  $X_{\mathcal{D}_j}$  follows a Pareto distribution with the parameters  $\alpha$  and  $c$ . Hence,

$$\begin{aligned} &\mathbb{E}_{X_{\mathcal{D}_p} \sim \mathcal{P}_{mass}^p, X_{\mathcal{D}} \sim \mathcal{P}_{mass}} [-X_{\mathcal{D}_p} \log X_{\mathcal{D}}] \\ &= - \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}} \cdot \frac{x_1 + x_2 + x_3}{3} \cdot \log \frac{\sum_{i \in \{p,a,t\}} |\mathcal{D}_i| x_i}{\sum_{i \in \{p,a,t\}} |\mathcal{D}_i|} dx_1 dx_2 dx_3 \\ &= - \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha} (x_1 + x_2 + x_3)}{3 \prod_{i=1}^3 x_i^{\alpha+1}} \log \frac{\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3}{k + l + 1} dx_1 dx_2 dx_3 \\ &\mathbb{E}_{X_{\mathcal{D}_a} \sim \mathcal{P}_{mass}^a, X_{\mathcal{D}} \sim \mathcal{P}_{mass}} [-X_{\mathcal{D}_a} \log X_{\mathcal{D}}] \\ &= - \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha} x_2}{\prod_{i=1}^3 x_i^{\alpha+1}} \log \frac{\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3}{k + l + 1} dx_1 dx_2 dx_3 \end{aligned}$$

where  $j = 1, 2, 3$ . Therefore,  $\frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl}$  and  $\frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl}$  can be written as:

$$\frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} = -\frac{dS_1}{dl}, \quad \frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl} = -\frac{dS_2}{dl} \quad (5)$$

where

$$\begin{aligned} S_1 &= \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha} (x_1 + x_2 + x_3)}{3 x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \log \frac{\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3}{k + l + 1} dx_1 dx_2 dx_3 \\ S_2 &= \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \log \frac{\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3}{k + l + 1} dx_1 dx_2 dx_3. \end{aligned}$$

Proving that  $\frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl} = \Theta\left(-k \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl}\right)$  is equivalent to proving:

$$\lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} + \frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl}}{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl}} = 0, \quad (6)$$

By substituting (5) into (6), we have

$$\lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} + \frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl}}{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl}} = \frac{\lim_{k \rightarrow +\infty, l \rightarrow 0} \left(\frac{k \cdot dS_1 + dS_2}{dl}\right)}{\lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{k \cdot dS_1}{dl}}. \quad (7)$$

Now calculate the values of  $\frac{dS_1}{dl}$  for the case when  $l \rightarrow 0$  and  $k$  is sufficiently large.

$$\begin{aligned}
& \lim_{l \rightarrow 0} \frac{dS_1}{dl} \\
&= \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{(x_1 + x_2 + x_3)}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \\
& \quad \cdot \frac{(k+1+l)x_3 - (\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)}{(k+1+l)(\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)} dx_1 dx_2 dx_3 \\
&= \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{(x_1 + x_2 + x_3)x_3}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)} dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{(x_1 + x_2 + x_3)}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (k+1+l)} dx_1 dx_2 dx_3 \\
&= \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \lim_{l \rightarrow 0} \frac{(x_1 + x_2 + x_3)x_3}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)} dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (k+1+l)} dx_1 dx_2 dx_3 \\
&= \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1 + x_2 + x_3}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \\
& \quad \cdot \lim_{l \rightarrow 0} \left[ \frac{x_3}{\frac{k}{3}(x_1 + x_2 + x_3) + x_2} - \frac{x_3^2}{[\frac{k}{3}(x_1 + x_2 + x_3) + x_2]^2} l \right] dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (k+1+l)} dx_1 dx_2 dx_3 \tag{8}
\end{aligned}$$

$$\begin{aligned}
&= \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1 + x_2 + x_3}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \frac{x_3}{\frac{k}{3}(x_1 + x_2 + x_3) + x_2} dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (k+1+l)} dx_1 dx_2 dx_3 \\
&= \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1 + x_2 + x_3}{3x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \left[ \frac{x_3}{(x_1 + x_2 + x_3)} \frac{3}{k} - \frac{x_2 x_3}{(x_1 + x_2 + x_3)^2} \frac{9}{k^2} \right] dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1} (k+1+l)} dx_1 dx_2 dx_3 \tag{9}
\end{aligned}$$

$$\begin{aligned}
&= \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \frac{1}{k(k+1)} dx_1 dx_2 dx_3 \\
& \quad - \alpha^3 c^{3\alpha} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \frac{3x_2 x_3}{k^2(x_1 + x_2 + x_3)} dx_1 dx_2 dx_3 \tag{10}
\end{aligned}$$

Here, we apply a Taylor expansion to Equation (8) and Equation (9) with respect to  $l$  and  $\frac{1}{k}$ , respectively.

Similarly, we can calculate the value of  $\frac{dS_2}{dl}$ .

$$\begin{aligned}
& \lim_{l \rightarrow 0} \frac{dS_2}{dl} \\
&= \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_2}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \\
& \quad \cdot \frac{(k+1+l)x_3 - (\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)}{(k+1+l)(\frac{k}{3}(x_1 + x_2 + x_3) + x_2 + lx_3)} dx_1 dx_2 dx_3 \\
&= \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \cdot \left( \frac{3x_2 x_3}{k(x_1 + x_2 + x_3)} - \frac{1}{k+1+l} \right) dx_1 dx_2 dx_3 \tag{11}
\end{aligned}$$

As a result, Equation (7) can be written as:

$$\begin{aligned}
& \lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} + \frac{d\gamma_{p\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl}}{k \cdot \frac{d\gamma_{p\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl}} \\
&= \lim_{k \rightarrow +\infty, l \rightarrow 0} \left( \frac{k \cdot dS_1 + dS_2}{d l} \right) \\
&= \frac{\lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{k \cdot dS_1}{d l}}{\lim_{k \rightarrow +\infty, l \rightarrow 0} \left( k \cdot \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}} \left[ \frac{x_1}{k(k+1)} - \frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)} \right] dx_1 dx_2 dx_3 \right. \\
&\quad \left. + \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}} \cdot \left( \frac{3x_2x_3}{k(\sum_{i=1}^3 x_i)} - \frac{1}{k+1+l} \right) dx_1 dx_2 dx_3 \right)} \\
&= \frac{\lim_{k \rightarrow +\infty, l \rightarrow 0} k \cdot \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{1}{\prod_{i=1}^3 x_i^{\alpha+1}} \left[ \frac{x_1}{k(k+1)} - \frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)} \right] dx_1 dx_2 dx_3}{\lim_{k \rightarrow +\infty, l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\alpha^3 c^{3\alpha}}{\prod_{i=1}^3 x_i^{\alpha+1}} \cdot \frac{l}{(k+1)(k+1+l)}} \\
&= \frac{\lim_{k \rightarrow +\infty, l \rightarrow 0} k \cdot \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{1}{\prod_{i=1}^3 x_i^{\alpha+1}} \left[ \frac{x_1}{k(k+1)} - \frac{3x_2x_3}{k^2(\sum_{i=1}^3 x_i)} \right] dx_1 dx_2 dx_3}{=0}.
\end{aligned} \tag{12}$$

Next, we prove that  $\frac{d\gamma_{p\theta}^{\mathcal{D}_1/\mathcal{D}}}{dl} < 0$ . For sufficiently large  $k$ , since

$$\begin{aligned}
& \lim_{l \rightarrow 0} \frac{dS_1}{d l} \\
&= \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{x_1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \frac{1}{k(k+1)} dx_1 dx_2 dx_3 \\
&\quad - \alpha^3 c^{3\alpha} \lim_{l \rightarrow 0} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{1}{x_1^{\alpha+1} x_2^{\alpha+1} x_3^{\alpha+1}} \frac{3x_2x_3}{k^2(x_1+x_2+x_3)} dx_1 dx_2 dx_3 \\
&= \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{\frac{k}{k+1} x_1(x_1+x_2+x_3) - 3x_1x_3}{k(x_1+x_2+x_3)x_1^{\alpha+1}x_2^{\alpha+1}x_3^{\alpha+1}} dx_1 dx_2 dx_3 \\
&:= \frac{1}{k} \int_c^{+\infty} \int_c^{+\infty} \int_c^{+\infty} \frac{f(x_1, x_2, x_3)}{g(x_1, x_2, x_3)} dx_1 dx_2 dx_3.
\end{aligned} \tag{13}$$

For  $c < u < v < w$ , let

$$\begin{aligned}
\hat{f} &= x_1(x_1+x_2+x_3) - 3x_1x_3 = x_1(x_1+x_2-2x_3), \\
h(u, v, w) &= \sum_{\pi \in S_3} f(\pi(u, v, w)), \\
\hat{h}(u, v, w) &= \sum_{\pi \in S_3} \hat{f}(\pi(u, v, w)).
\end{aligned} \tag{14}$$

Here,  $S_3$  denotes the symmetric group of degree 3 and  $\pi \in S_3$  denotes a permutation in  $S_3$ . Observe that

$$\begin{aligned}
& \lim_{k \rightarrow +\infty} h(u, v, w) = \hat{h}(u, v, w)^-, \\
& \hat{h}(u, v, w) = \sum_{\pi \in S_3} \hat{f}(\pi(u, v, w)) \\
&= \sum_{\substack{\pi \in S_3 \\ (u_0, v_0, w_0) = \pi(u, v, w)}} u_0(u_0 + v_0 - 2w_0) \\
&= \sum_{\substack{\pi \in S_3 \\ (u_0, v_0, w_0) = \pi(u, v, w)}} \frac{1}{2}(u_0 - v_0)^2 > 0.
\end{aligned} \tag{15}$$